Question:

A 34-year-old woman presents to her OB/GYN with complaints of missing her last 3 periods as well as intermittent spontaneous milky-white nipple discharge bilaterally for the past 3 months. Vital signs are stable and within normal limits. Neurologic examination is without abnormality, including normal visual fields. Serology and MRI of the brain are ordered, with results pending. Which of the following sets of laboratory results would be expected in this patient?

Answer Choices:

A. Decreased prolactin, decreased FSH, decreased LH

B. Decreased prolactin, increased FSH, increased LH

C. Increased prolactin, decreased FSH, increased LH

D. Increased prolactin, decreased FSH, decreased LH

Reasoning + Answer:

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The patient is a 34-year-old woman presenting with amenorrhea (absence of menstruation) and bilateral milk discharge (galactorrhea). These symptoms are classically associated with hyperprolactinemia, which can suppress the hypothalamic-pituitary-ovarian axis.

Hyperprolactinemia leads to:

- Inhibition of gonadotropin-releasing hormone (GnRH), which results in decreased secretion of luteinizing hormone (LH) and follicle-stimulating hormone (FSH).
- Elevated prolactin levels, which directly cause galactorrhea and amenorrhea.

Thus, in a patient with hyperprolactinemia, the expected laboratory findings are:

- Increased prolactin
- Decreased FSH
- Decreased LH

This fits with option D.

Note: Although the MRI and serology are pending, the clinical presentation is consistent with a pituitary disorder (e.g., prolactinoma), which would cause elevated prolactin and suppressed gonadotropins.

Option D accurately reflects this pattern.

<answer>Increased prolactin, decreased FSH, decreased LH</answer>

<tool_call> reasoning process here<tool_call><answer>Increased prolactin, decreased FHS, decreased LH</answer>